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STAT 341 Spring 2025

Computational Statistics and Data Analysis

Section 001, 101

Class Schedule

Course	Meet Days	Meet Time	Location	Instructor(s)
STAT 341 001 [LEC]	Mon, Wed <i>May 5 - Jul 30</i>	10:00AM - 11:20AM	AL 113	Michael John Davis mj2davis@uwaterloo.ca
STAT 341 101 [TUT]	Fridays <i>May 16 - Jul 30</i>	02:30PM - 03:20PM	DC 1351	Michael John Davis mj2davis@uwaterloo.ca

The following make-up days have been scheduled for Spring 2025:

- **Mon May 19** (Victoria Day) > **Wed July 2** (Monday Schedule)
- **Mon June 30** (Additional Day) > **Mon July 28**
- **Tue July 1** (Canada Day) > **Tue Jul 29**

The loss of a Wednesday class on **July 2** due to the Monday schedule will be made up on **Wed July 30**.

schedule data automatically refreshed daily

Instructional Team

Lecturer: Dr. Jack Davis - mj2davis@uwaterloo.ca , M3 2108

ISC: Pam Bandara - pbandara@uwaterloo.ca

For any questions not related to course content, please contact Pam. This includes requests for accommodations (e.g., due to illness), re-mark inquiries, concerns about academic performance, and issues related to LEARN, Crowdmark, or Piazza, as well as any personal matters.

Course Description

Calendar Description for STAT 341

A computationally focused approach to statistical reasoning in the context of real data. Functional programming in R and algorithms will be used to define interesting attributes of finite populations and their sampling characteristics. Computational approaches to inductive inference and the assessment of predictive accuracy.

[View requirements for STAT 341 ^{\(1\)}](#)

Learning Outcomes

By the end of this course students should be able to...

Understand the fundamentals of population-based statistical analyses, as well as implicitly/ explicitly defined population attributes.

Understand the basics of sampling and the underlying ideas/ mechanisms of probability sampling and related issues.

Understand inductive inference including target and study populations, measurements, comparing subpopulations, random intervals and coverage probabilities, and resampling methods such as the bootstrap.

Understand the fundamentals of prediction and how its accuracy is measured.

Generate high-quality visualizations and inferential analyses using R for the topics above.

Prepare reports that communicate your statistical analyses using LaTeX and R Markdown.

Tentative Class Plan

The following outline is tentative and may shift a bit. The dates are listed for the section with the Tuesday/Thursday schedule. For the Monday/Wednesday schedule subtract one day for each lecture.

Week	Week Start	Class Topic	Notes	Deliverables
1	2025-05-05	Intro, population attributes		
2	2025-05-12	graphical attributes, power transforms		
3	2025-05-19	Implicit attributes, robust regression	M Off, M1 Tut	A1 due Fri
4	2025-05-26	Gradient descent		
5	2025-06-02	Newton-Raphson, IRLS		M1 Fri 4:30
6	2025-06-09	Sampling	M2 Tut	A2 due Fri

7	2025-06-16	Sampling, Inclusion Prob		
8	2025-06-23	Inductive Inference		M2 Fri 4:30
9	2025-06-30	T-like measure, Multiple testing	M+T Off	A3 due Fri
10	2025-07-07	Intervals, bootstrap		
11	2025-07-14	Extensions to bootstrap		
12	2025-07-21	Prediction accuracy	Final Tut	
13	2025-07-28	Makeup Days M + T + W	Final Tut	A4 due Wed

Required Materials & Technologies

Note: Any prices provided in course outlines are best estimates based on recent online prices and do not include shipping or taxes. Prices may vary between retailers.

This course has **no additional costs** for students.

Readings

Only the notes will be needed. The old notes are 90% of the notes that we will use, and the 10% will be niche topics and assignment-specific.

Technology

Name of Technology	Notes / Comments	Required	URL (student access)	Price (CAD)
R, Rstudio		Required		0
LEARN		Required		0
Piazza		Recommended		0

Assessments & Activities

Component / Activity	Date or Due Date	Location / Submission Method	Weight (%)
Assignment 1	Friday, May 23 11:59pm	Crowdmark	5
Assignment 2	Friday, June 13 11:59pm	Crowdmark	5
Assignment 3	Friday, July 4, 11:59pm	Crowdmark	5

Assignment 4	Friday July 25, 11:59pm	Crowdmark	5
Midterm 1	Friday, June 6, 4:30 - 5:50pm	DC 1350, DC 1351	20
Midterm 2	Friday, June 27, 4:30 - 5:50pm	DC 1350, DC 1351	20
Final	TBA	TBA	40

Assignments

There are 4 assignments due on Fridays at 11:59pm, and each is released approximately 7 days before the deadline. Assignments will be submitted to Crowdmark. If you choose to type your submission, you can use word with its equation editor and then save it as pdf, or use LaTeX, whichever you are more comfortable with.

Midterms

There are two midterms held during tutorial. Each will have duration of 80 minutes . Full details for each test will be posted on LEARN.

Late / Missed Content

20% penalty per day, up to 2 days late. Email the ISC Pam Bandara pbandara@uwaterloo.ca in advance if you need accommodations.

Assignment Screening

No automated assignment screening will be utilized within this course.

Generative AI

Generative artificial intelligence (GenAI) trained using large language models (LLM) or other methods to produce text, images, music, or code, like Chat GPT, DALL-E, or GitHub CoPilot, **may be used under specific conditions** in this course with proper documentation, citation, and acknowledgement. **Permitted uses of and expectations for using GenAI will discussed in class and outlined on assignment instructions.**

[Recommendations for how to cite generative AI in student work at the University of Waterloo may be found through the Library. ^{\(2\)}](#)

Please be aware that generative AI is known to falsify references to other work and may fabricate facts and inaccurately express ideas. GenAI generates content based on the input of other human authors and may therefore contain inaccuracies or reflect biases.

To protect the privacy and security of any data entered, students should use the University's version of [Co-Pilot ^{\(3\)}](#) and login with their UW ID. Data entered into other systems can be added to training sets, monitored, geolocated and even reproduced as output which may share private personal information or result in intellectual property breaches.

In addition, you should be aware that the legal/copyright status of generative AI inputs and outputs is unclear. Exercise caution when using large portions of content from AI sources, especially images. [More information is available from the Copyright Advisory Committee.](#) (4)

You are accountable for the content and accuracy of all work you submit in this class, including any supported by generative AI. You should be able to readily demonstrate your knowledge of your submissions. To demonstrate your learning, you should keep your rough notes, including sources, research notes, brainstorming, drafting notes and prompts. You may be asked to submit these notes along with earlier drafts of your work, either through saved drafts or saved versions of a document.

Notice of Recording

Activities for this course involve recording, in partial fulfillment of the course learning outcomes. You will receive notification of recording via at least one of the following mechanisms: within the Learning Management System (LEARN), a message from your course instructor, course syllabus/website, or other means. Some technologies may also provide a recording indicator. Images, audio, text/chat messaging that have been recorded may be used and/or made available by the University to *[insert to whom¹]* for the purpose of *[insert purpose²]*. Recordings will be managed according to the University records classification scheme, [WatClass](#) (5), and will be securely destroyed when no longer needed by the University. Your personal information is protected in accordance with the [Freedom of Information and Protection of Privacy Act](#) (6), as well as [University policies and guidelines](#) (7) and may be subject to disclosure where required by law.

The University will use reasonable means to protect the security and confidentiality of the recorded information, but cannot provide a guarantee of such due to factors beyond the University's control, such as recordings being forwarded, copied, intercepted, circulated, disclosed, or stored without the University's knowledge or permission or the introduction of malware into computer system which could potentially damage or disrupt the computer, networks, and security settings. The University is not responsible for connectivity/technical difficulties or loss of data associated with your hardware, software or Internet connection.

By engaging in course activities that involve recording, you are consenting to the use of your appearance, image, text/chat messaging, and voice and/or likeness in the manner and under the conditions specified herein. (In the case of a live stream event, if you choose not to have your image or audio recorded, you may [disable the audio and video functionality](#) (8). Instructions to participate using a pseudonym instead of your real name are included where the feature exists; however, you must disclose the pseudonym to your instructor in advance in order to facilitate class participation.) If you choose not to be recorded, this notice serves as confirmation of your understanding that *[indicate alternative action that students can take³]*.

You are not permitted to disclose the link to/URL of an event or an event session recording or copies of recording to anyone, for any reason. Recordings are available only to authorized individuals who have been directly provided the above instructions/link for their use. Recordings for personal use, required to facilitate your learning and preparation of personal course/lecture notes, should not be shared with others without the permission of the instructor or event coordinator. Review the University's [guidelines for faculty, staff and students entering relationships with external organizations offering access to course materials](#) (9) for more information on your obligations with respect to keeping copies of course materials. For more information about accessibility, connect with [AccessAbility Services](#) (10).

Administrative Policy

1. **Remark Requests:** You may appeal the way an assignment/midterm was marked by submitting it to the ISC Pam Bandara pbandara@uwaterloo.ca within 1 week of it being returned. Your email must include a completed Remark Request Form. Please find the form and full details of the process posted on LEARN.
2. **Late Penalties:** A late penalty of **20% per day late, up to a maximum of two days** will be applied to assignments for late submissions to Crowdmark. Please do not email us if you have technical issues that cause your submission to be a few minutes late. No waiving of this penalty will occur for technical issues. Instead, we encourage you to plan in advance and submit your assignments earlier than the moment of the deadline.
3. **Illness:** If you are ill, you must submit a VIF indicating severe illness. (If you are self-isolating due to flu-like or COVID-19 symptoms, you can instead complete the self-declaration form in Quest). Details can be found here: <https://uwaterloo.ca/math/accommodations> ⁽¹¹⁾
 - If you are ill for a midterm, you must provide satisfactory documentation within 48 hours by emailing an electronic copy of the VIF to the Instructional Support Coordinator (ISC) as well as submitting to the MUO (or email the ISC in addition to completing the self-declaration). For satisfactory documentation, the weight of the test will be shifted to the final exam.
 - If you are ill for the final exam, you will receive an INC grade and be required to complete the exam during the next course offering, if you are achieving a passing grade going into the exam. Further, you must have written at least one of the two midterms to receive an INC. If not, a failing grade of DNW will be assigned.
4. If at any time you are feeling sick, or have been directed by public health to self-isolate, you should not be attending class. We encourage you to get the lecture notes from a classmate or from a recording on Learn, but if needed please email your instructor directly and we will determine an appropriate accommodation for the situation.

Mental Health Support

The Faculty of Math encourages students to seek out mental health support if needed.

On-campus Resources:

- Campus Wellness <https://uwaterloo.ca/campus-wellness/> ⁽¹²⁾
- Counselling Services: counselling.services@uwaterloo.ca/ 519-888-4567 ext 32655
- MATES: one-to-one peer support program offered by Federation of Students (FEDS) and Counselling Services: mates@uwaterloo.ca
- Health Services: located across the creek from the Student Life Centre, 519-888-4096.

Off-campus Resources:

- Good2Talk (24/7): Free confidential help line for post-secondary students. Phone: 1-866-925-5454
- Here 24/7: Mental Health and Crisis Service Team. Phone: 1-844-437-3247
- OK2BME: set of support services for lesbian, gay, bisexual, transgender, or questioning teens in Waterloo. Phone: 519-884-0000 extension 213

Personal Mental Health Support Message

You're not alone. Lots of people look like they have it together in public and online when they're a mess inside too. You're not the only student facing what you're facing, and it really is okay to seek out help from the above sources if you need.

Getting help is not an admission of failure. For the same reason that you cannot walk off a broken leg or a heart attack. It's just a thing that some people need to do.

It probably gets better. Obviously, nobody can guarantee that, but it is extremely likely to get better. More specifically, you get stronger and more capable at dealing with the problems in your life.

You are not your grades. Grades are not a reflection of your worth as a human being. They got you into university, so they've already done most of the work, but at this point, the most they can really do for you decide if you can get into graduate school before doing industry experience. (And if you're in the graduate version of this class, then congrats! You barely need grades at all now.)

This too will pass. Some people will say that being a student was the best time of their life. Don't let that be your reason for thinking this is as good as it gets. Everyone's life path is different.

Every day's 'best' is a little different. If you could do your theoretical best every day, then it wouldn't be your best it would be your average.

Others are not yardsticks. The takeaway message is NOT 'if they can become a prof with a mental illness, then I should be able to as well.', please don't use others as a point of comparison like that.

Take care of yourself when you can, don't blame yourself when you can't.

The world is better with you in it.

Diversity and AccessAbility

It is my intent that students from all diverse backgrounds and perspectives be well served by this course, and that students' learning needs be addressed both in and out of class. We recognize the immense value of the diversity in identities, perspectives, and contributions that students bring, and the benefit it has on our educational environment. Your suggestions are encouraged and appreciated. Please let us know ways to improve the effectiveness of the course for you personally or for other students or student groups. In particular:

We will gladly honour your request to address you by an alternate/preferred name or gender pronoun. Please advise us of this preference early in the semester so I may make appropriate changes to the records.

We will honour your religious holidays and celebrations. Please inform me of these at the start of the course.

We will follow AccessAbility Services guidelines and protocols on how to best support students with different learning needs. If you require academic accommodations to lessen the impact of your disability, please register with AccessAbility at the beginning of each academic term

Intellectual Property

Students should be aware that this course contains the intellectual property of their instructor, TA, and/or the University of Waterloo, including items such as:

- Lecture content, spoken and written (and any audio/video recording thereof)
- Lecture handouts, presentations, and other materials prepared for the course
- Questions or solution sets from any type of assessment
- Work protected by copyright (e.g., any work used by the instructor with permission of the copyright owner).

Sharing this intellectual property without the owner's permission is a violation of intellectual property rights. For this reason, it is necessary to ask the instructor, TA and/or the University of Waterloo for permission before uploading and sharing the intellectual property of others online (e.g., to an online repository). Permission is also necessary before sharing the intellectual property of others from completed courses with students taking the same/similar courses in subsequent terms/years.

Please alert the instructor if you become aware of intellectual property belonging to others (past or present) circulating, either through the student body or online. In many cases, instructors might be happy to allow distribution of certain materials. However, doing so without expressed permission is considered a violation of intellectual property rights.

Territorial Acknowledgement: The University of Waterloo acknowledges that much of our work takes place on the traditional territory of the Neutral, Anishinaabeg, and Haudenosaunee peoples. Our main campus is situated on the Haldimand Tract, the land granted to the Six Nations that includes six miles on each side of the Grand River. Our active work toward reconciliation takes place across our campuses through research, learning, teaching, and community building, and is co-ordinated within the [Office of Indigenous Relations](#). ⁽¹³⁾

University Policy

Mental Health: At the University of Waterloo, we are dedicated to supporting your mental and emotional well-being. Our Counselling Services offer confidential support, including individual counselling, workshops, and crisis intervention.

If you're struggling, please reach out for help at 519-888-4096 or visit [their website](#) ⁽¹⁴⁾ for more information.

Academic integrity: In order to maintain a culture of academic integrity, members of the University of Waterloo community are expected to promote honesty, trust, fairness, respect and responsibility. [Check [the Office of Academic Integrity](#) ⁽¹⁵⁾ for more information.]

Grievance: A student who believes that a decision affecting some aspect of their university life has been unfair or unreasonable may have grounds for initiating a grievance. Read [Policy 70, Student Petitions and Grievances](#),

[Section 4 \(16\)](#). When in doubt, please be certain to contact the department's administrative assistant who will provide further assistance.

Discipline: A student is expected to know what constitutes academic integrity to avoid committing an academic offence, and to take responsibility for their actions. [Check [the Office of Academic Integrity \(17\)](#) for more information.] A student who is unsure whether an action constitutes an offence, or who needs help in learning how to avoid offences (e.g., plagiarism, cheating) or about "rules" for group work/collaboration should seek guidance from the course instructor, academic advisor, or the undergraduate associate dean. For information on categories of offences and types of penalties, students should refer to [Policy 71, Student Discipline \(18\)](#). For typical penalties, check [Guidelines for the Assessment of Penalties \(19\)](#).

Appeals: A decision made or penalty imposed under [Policy 70, Student Petitions and Grievances \(20\)](#) (other than a petition) or [Policy 71, Student Discipline \(21\)](#) may be appealed if there is a ground. A student who believes they have a ground for an appeal should refer to [Policy 72, Student Appeals \(22\)](#).

Note for students with disabilities and disabling conditions: The University of Waterloo recognizes its obligations under the Ontario Human Rights Code to accommodate students with known or suspected disabilities and disabling conditions (e.g. medical conditions, injuries, impacts of trauma such as from violence or discrimination) to the point of undue hardship. To support this obligation, [AccessAbility Services \(23\)](#) (AAS) collaborates with all academic departments and schools to facilitate academic accommodations for students with disabilities and disabling conditions without compromising the academic integrity of the curriculum. If you believe you may require academic accommodations (e.g., testing accommodations, classroom accommodations), register with AAS as early in the term as possible by completing the [online application \(24\)](#). Students already registered with AAS must activate their accommodations for each of their courses at the beginning of each term using AAS' online system. If you require assistance, contact AAS by phone (519-888-4567 ext. 35082), email (access@uwaterloo.ca) or in-person (Needles Hall North, 1st Floor, Room 1401).

Turnitin.com: Text matching software (Turnitin®) may be used to screen assignments in this course. Turnitin® is used to verify that all materials and sources in assignments are documented. Students' submissions are stored on a U.S. server, therefore students must be given an alternative (e.g., scaffolded assignment or annotated bibliography), if they are concerned about their privacy and/or security. Students will be given due notice, in the first week of the term and/or at the time assignment details are provided, about arrangements and alternatives for the use of Turnitin in this course.

It is the responsibility of the student to notify the instructor if they, in the first week of term or at the time assignment details are provided, wish to submit alternate assignment.

Reference: Links from Document

1. <https://acal.fast.uwaterloo.ca/course/1255/STAT/341>
2. https://subjectguides.uwaterloo.ca/chatgpt_generative_ai
3. <https://copilot.microsoft.com/>
4. <https://uwaterloo.ca/copyright-at-waterloo/teaching/generative-artificial-intelligence>

5. <https://uwaterloo.ca/records-management/records-classification-and-retention-schedules>
6. <https://www.ontario.ca/laws/statute/90f31>
7. <https://uwaterloo.ca/privacy/>
8. <https://uwaterloo.ca/student-it-services/>
9. <https://uwaterloo.ca/secretariat/faculty-staff-and-students-entering-relationships-external>
10. <https://uwaterloo.ca/accessability-services/>
11. <https://uwaterloo.ca/math/accommodations>
12. <https://uwaterloo.ca/campus-wellness/>
13. <https://uwaterloo.ca/indigenous>
14. <https://uwaterloo.ca/students/health-and-well-being/counselling-appointments>
15. <https://uwaterloo.ca/academic-integrity/>
16. <https://uwaterloo.ca/secretariat/policies-procedures-guidelines/policy-70>
17. <https://uwaterloo.ca/academic-integrity/>
18. <https://uwaterloo.ca/secretariat/policies-procedures-guidelines/policy-71>
19. <https://uwaterloo.ca/secretariat/guidelines/guidelines-assessment-penalties>
20. <https://uwaterloo.ca/secretariat/policies-procedures-guidelines/policy-70>
21. <https://uwaterloo.ca/secretariat/policies-procedures-guidelines/policy-71>
22. <https://uwaterloo.ca/secretariat/policies-procedures-guidelines/policy-72>
23. <https://uwaterloo.ca/accessability-services/>
24. <https://uwaterloo.ca/accessability-services/students/applying-academic-accommodations/documentation-information-forms>

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